

Tandberg TDV-2200 ND Terminal Analysis - Updated with Specifications

Updated Analysis Based on ND Terminal Escape Sequence Specification

After analyzing the ND terminal specification document, I can now provide a much more accurate picture of the Tandberg TDV-2200's escape sequence processing capabilities.

Updated Function Names and Analysis

Core ND Terminal Processing Functions

Address	Original Name	Updated Name	ND Function
0x1BE3	process_incoming_character	nd_character_processor	Main ND character processing dispatcher
0x19CE	call_escape_sequence_handler	nd_escape_sequence_dispatcher	ND escape sequence function dispatcher
0x1F69	parse_hex_parameter_sequence	nd_parse_csi_parameters	Parse CSI numeric parameters
0x1F85	validate_hex_digit_input	nd_validate_parameter_digit	Validate parameter digits (0-9, A-F)
0x11CE	handle_special_control_sequence	nd_handle_control_sequence	Handle ND control sequences

Updated Memory Variables

Address	Original Name	Updated Name	ND Purpose
0x5E08	escape_sequence_handler_pointer	nd_escape_handler_function_pointer	Points to active ND escape handler
0x5C2A	serial_function_pointer	nd_character_processing_function_pointer	Main ND character processor
0x5E20	character_generator_select_flag	nd_character_set_selection_flag	ND alternative character set selection

ND Character Code Mappings

ND Private Control Codes Found in Firmware

Code	Hex	ND Function	Description
252	0xFC	ND_ESC_TRIGGER	Triggers ND escape sequence processing
206	0xCE	ND_PARAM_MODE	Enters parameter parsing mode for CSI sequences
207	0xCF	ND_EXT_PARAM_MODE	Extended parameter parsing (multi-character)
177	0xB1	ND_CTRL_SEQ_1	ND private control sequence 1
178	0xB2	ND_CTRL_SEQ_2	ND private control sequence 2
179	0xB3	ND_CTRL_SEQ_3	ND private control sequence 3
180	0xB4	ND_CTRL_SEQ_4	ND private control sequence 4
184	0xB8	ND_SPECIAL_CTRL	Special ND control function

ND Escape Sequences Supported

ND Private Escape Sequences (ESC n)

Based on the specification, the terminal supports:

Sequence	Code	Mnemonic	Function
ESC 1	0x31	NDSS1	Single shift to alternative character set 1
ESC 2	0x32	NDSS2	Single shift to alternative character set 2
ESC 3	0x33	NDSS3	Single shift to alternative character set 3
ESC 4	0x34	NDSS4	Single shift to alternative character set 4
ESC 5	0x35	NDSS5	Single shift to alternative character set 5
ESC 6	0x36	NDSS6	Single shift to alternative character set 6
ESC 7	0x37	NDSC	Save cursor position
ESC 8	0x38	NDRC	Restore cursor position
ESC 9	0x39	NDSS7	Single shift to alternative character set 7
ESC :	0x3A	NDSS8	Single shift to alternative character set 8
ESC ;	0x3B	NDSS9	Single shift to alternative character set 9
ESC =	0x3D	NDEAKM	Enter alternate keypad mode
ESC >	0x3E	NDXAKM	Exit alternate keypad mode

Standard C1 Control Sequences

Sequence	Code	Mnemonic	Function
ESC D	0x44	IND	Index (cursor down with scroll)
ESC E	0x45	NEL	Newline (CR+LF)
ESC H	0x48	HTS	Horizontal tab set
ESC M	0x4D	RI	Reverse index (cursor up with scroll)
ESC [	0x5B	CSI	Control sequence introducer
ESC c	0x63	RIS	Reset to initial state

ND Private CSI Sequences Found

CSI Sequences in Firmware Memory

The firmware contains these ND private CSI sequences:

CSI Code	Decimal	Mnemonic	Function
CSI 112 p	0x70	NDLIWA	Insert lines in work area
CSI 113 q	0x71	NDDLWA	Delete lines in work area
CSI 114 r	0x72	NDSTBM	Set top and bottom margin
CSI 115 s	0x73	NDICHE	Insert characters with extent
CSI 116 t	0x74	NDDCHE	Delete characters with extent
CSI 117 u	0x75	NDSREC	Save rectangle
CSI 118 v	0x76	NDRREC	Restore rectangle
CSI 119 w	0x77	NDSSKL	Set soft-key level
CSI 120 x	0x78	NDREQ	Request terminal parameters
CSI 121 y	0x79	NDTST	Invoke confidence test
CSI 122 z	0x7A	NDSAR	Set attribute in rectangle

Additional CSI Sequences Found

The firmware also contains many other CSI sequences (0-87) suggesting support for:

- Cursor positioning and movement
- Character and line insertion/deletion
- Screen region manipulation
- Attribute setting and control
- Function key programming

ND Terminal Architecture

Character Processing Flow

c

```
nd_character_processor(uint8_t character) {
    switch (character) {
        case 0xFC: // ND escape trigger
            nd_escape_sequence_dispatcher();
            break;

        case 0xCE: // Parameter mode
            nd_parse_csi_parameters();
            break;

        case 0xCF: // Extended parameter mode
            // Read multiple characters for complex sequences
            do {
                character = try_read_serial_data();
            } while (parsing_needed);
            break;

        case 0xB1: // ND control sequence 1
        case 0xB2: // ND control sequence 2
        case 0xB3: // ND control sequence 3
        case 0xB4: // ND control sequence 4
            nd_handle_control_sequence(character);
            break;

        default:
            if (character >= 0x20 && character <= 0x7F) {
                // Normal printable characters
                output_character_to_screen(character);
            }
            break;
    }
}
```

ND Function Pointer Architecture

c

```
// Main character processing dispatcher
(*nd_character_processing_function_pointer)(); // 0x5C2A

// Escape sequence handler dispatcher
(*nd_escape_handler_function_pointer)(); // 0x5E08
```

ND Terminal Capabilities

Supported ND Features

1. **Alternative Character Sets:** 9 alternative character sets (NDSS1-NDSS9)
2. **Cursor Control:** Save/restore cursor (NDSC/NDRC)
3. **Keypad Modes:** Alternate keypad mode (NDEAKM/NDXAKM)
4. **Rectangle Operations:** Save/restore/manipulate screen rectangles
5. **Work Area Control:** Insert/delete lines and characters in work areas
6. **Margin Control:** Set top and bottom margins (NDSTBM)
7. **Soft Keys:** Programmable function keys (NDSSKL)
8. **Diagnostics:** Built-in confidence tests (NDTST)
9. **Parameter Reporting:** Terminal parameter requests (NDREQ)

Character Set Support

- **G0/G1 Character Sets:** Standard and alternate character sets
- **Single Shift:** Temporary character set switching
- **Locking Shift:** Permanent character set switching
- **Alternative Character Sets:** 9 different alternative sets

Emulator Implementation Guide

Key ND Character Codes to Handle

c

```
#define ND_ESC_TRIGGER      0xFC    // Escape sequence trigger
#define ND_PARAM_MODE      0xCE    // Parameter parsing mode
#define ND_EXT_PARAM_MODE  0xCF    // Extended parameter parsing
#define ND_CTRL_SEQ_1      0xB1    // Control sequence 1
#define ND_CTRL_SEQ_2      0xB2    // Control sequence 2
#define ND_CTRL_SEQ_3      0xB3    // Control sequence 3
#define ND_CTRL_SEQ_4      0xB4    // Control sequence 4
```

Function Pointer Monitoring

c

```
// Monitor these function pointers for mode changes
uint16_t* nd_char_processor = (uint16_t*)0x5C2A;
uint16_t* nd_escape_handler = (uint16_t*)0x5E08;
```

CSI Sequence Processing

c

```
// CSI sequences use decimal parameters followed by underscore
// Format: ESC [ nnn _
// Where nnn is decimal parameter (000-255)
void handle_csi_sequence(uint8_t param) {
    switch (param) {
        case 112: nd_insert_lines_work_area(); break; // NDLIWA
        case 113: nd_delete_lines_work_area(); break; // NDDLWA
        case 114: nd_set_top_bottom_margin(); break; // NDSTBM
        case 115: nd_insert_chars_extent(); break; // NDICHE
        case 116: nd_delete_chars_extent(); break; // NDDCHE
        case 117: nd_save_rectangle(); break; // NDSREC
        case 118: nd_restore_rectangle(); break; // NDRREC
        case 119: nd_set_softkey_level(); break; // NDSSKL
        case 120: nd_request_parameters(); break; // NDREQ
        case 121: nd_invoke_confidence_test(); break; // NDTST
        case 122: nd_set_attribute_rectangle(); break; // NDSAR
        // ... handle other sequences
    }
}
```

Conclusion

The Tandberg TDV-2200 implements a **comprehensive ND terminal specification** with:

- **Full ND private escape sequence support**
- **Extensive CSI command set** (80+ sequences)
- **Multiple character set support** (9 alternative sets)
- **Advanced terminal features** (rectangles, work areas, soft keys)
- **Professional diagnostic capabilities**

This makes it a **high-end professional terminal** with capabilities far beyond basic VT100 compatibility, designed for sophisticated applications requiring advanced screen manipulation and character set support.